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SEAT No. _____

No. of Printed Pages : 3

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SARDAR PATEL UNIVERSITY**M.SC(IT) Sem-II (NC) EXAMINATION****WEDNESDAY, 20th MARCH, 2019****10.00 am to 12.00 noon****SUBJECT: MATHEMATICS-II (PS02FIIT02)**

Maximum Marks: 70

Q-1 Write the correct option in the answer book.

[10]

(1) The set $\{x \in \mathbb{N} : 1 < x < 75\}$ is:

- (a) finite (b) Infinite (c) Empty (d) none

(2) The number of elements in a set $\{x \in \mathbb{N} : 2x^2 - 1 = 0\}$ are:

- (a) 0 (b) 8 (c)
- $\pm 1/2$
- (d) 1

(3) Let $A = \{0, 1\}$, then A closed under:

- (a) multiplication (b) addition (c) division (d) subtraction

(4) $P(8, 4) =$ (a) 30 (b) 1680 (c) 1860 (d) 56(5) If Karl-Pearson's coefficients of skewness $S_k = 0$ then the frequency distribution is

- (a) Not Skewed (b) Positively skewed
-
- (c) Negatively skewed (d) None of these

(6) Every monoid are:

- (a) group (b) ring (c) semigroup (d) none

(7) De Morgan's Law:

- (a)
- $(A \cup B)^c = A^c \cap B^c$
- (b)
- $(A \cap B)^c = (A \cap B)^c$
- (c)
- $(A \cup B)^c = A \cap B$
- (d) None

(8) Which of the following is not the measure of Dispersion?

- (a) Range (b) Mode (c) Mean deviation (d) Standard deviation

(9) $\binom{n}{0} + \binom{n}{n} =$

- (a) n (b) 1 (c) 2 (d) 3

(10) The set $\{\mathbb{N}, +\}$ is:

- (a) group (b) ring (c) monoid (d) semigroup

Q:2 Answer the following in short. (Any Ten)

[20]

(1) Find the power set of a set $\{1, 2, 3\}$.

(2) Define range and quartile deviation.

(1)

(P.T.O.)

(3) A class consists 5 boys and 4 girls. Find the number of ways to select a 3 member committee with at least one girl.

(4) Let $A = \{2, 4, 6, \dots\}$. Is A closed under: (i) multiplication and (ii) addition?

(5) Define the terms: binary operation and group.

(6) Prove that $\binom{17}{6} = \binom{16}{5} + \binom{16}{6}$

(7) Find Karl Pearson's coefficient of skewness for the following distributions:

	Distribution A	Distribution B
Mean	100	90
Mode	90	80
S.D.	10	10

(8) Simplify : $(n+1)!/(n-1)!$

(9) If S is a nonempty set with the operation $a*b = a$. Is the operation * associative?

(10) If $A = \{1, 2, 3, \dots, 10\}$ and $B = \{5, 6, 7, \dots, 15\}$ then find the symmetrical difference between A and B.

(11) Explain the positive correlation with two examples.

(12) Find dual of the following: (i) $(A \cup B \cup C)^c = (A \cup C)^c \cap (A \cup B)^c$,

(ii) $(A \cup \phi) \cap (U \cup A^c) = A$.

Q-3 (a) Prove that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$.

[05]

(b) If $f(x) = x^2 + 5x$ and $g(x) = 3x + 2$ then find : (i) fog (ii) gof (iii) fog(a+1) (iv) gof(a+1) (v) fog(b) (vi) gof(2).

[05]

Q-3

OR

(c) Define a function F as $F(a, b) = 0$ if $a < b$.

[05]

$= F(a-b, b) + 1$ if $b \leq a$. Find $F(4, 3)$ and $F(14, 3)$.

(d) By using algebra of sets, prove that $(\phi \cup A) \cap (B \cup A) = A$.

[05]

Q-4 For a, b rational number set Q, define $a*b = ab/2$. Show that $(Q, *)$ is a group under

[05]

(a) binary operation *.

(b) For $a, b \in \mathbb{Q}$ (rational numbers), define $a * b = a + b + ab$. [05]

(i) Is $(\mathbb{Q}, *)$ Semigroup? (ii) Is $(\mathbb{Q}, *)$ Monoid? (iii) Find its inverse if it exist.

Q-4

OR

(c) Show that the set $\{1, 2, 3, 4, 5, 6\}$ is a group under multiplication modulo 7. [05]

(d) Prove that: [05]

(i) Left cancellation law holds in a group G .

(ii) $(ab)^{-1} = b^{-1}a^{-1}$

Q-5 How many diagonals has a regular polygon with n sides? Find the number of diagonals [05]

(a) in a octagon.

(b) Prove that $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$ [05]

Q-5

OR

(c) Find the number of ways that four mathematics books, three history books, three chemistry books and two sociology books can be arranged on a shelf so that all books of the same subject are together. [05]

(d) A student is to answer eight out of ten questions on an exam. Find the number m of ways that the student can chose eight questions. Also find m if the student must answer at least four out of the first five questions. [05]

Q-6 Calculate Karl Pearson's coefficient of Skewness for the following data: [05]

(a)

Size	1	2	3	4	5	6	7
Frequency	10	18	30	25	12	3	2

(b) Find the standard deviation for the following frequency distribution and obtain the coefficient of variance. [05]

x	6	7	8	9	10	11	12
f	3	6	9	13	8	5	4

Q-6

OR

In an experiment the number of grams of a given salt which dissolved in 100g of water was observed at eight different temperatures. The weights of salt at different temperatures are given below. [10]

Temp($^{\circ}\text{C}$)	0	10	20	30	40	50	60
Weight of salt(g)	51.1	61.5	67.2	72.6	73.5	82.2	83.2

Find the regression equation which could be used to predict the weight of salt given the temperature. Predict the weight of salt which would dissolve at temperatures (i) 25°C

(ii) 85°C .

— X —
(3)

